

Modern Geometry I & II: A Review

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Abstract

This is a review of the the Modern Geometry I and II notes by Nathaniel Miller published in the Journal of Inquiry-Based Learning in Mathematics. After describing the published notes, I will describe my teaching experience with using the notes as published, adapting the notes to a one semester course, student reactions to the notes, the strengths of these notes, and how using the notes has helped me to transition to a more IBL focused teaching in my own practice.

1 Introduction

This review will discuss my experience with the Miller Modern Geometry I and II notes [5, 6]. I am an Assistant Professor at Texas A&M University - Commerce with four years of IBL geometry teaching experience at two different institutions. I taught from the college geometry notes as written while

I was a graduate student at the University of Northern Colorado, and currently use a condensed version of the geometry notes for my one semester college geometry course at TAMUC.

This review has four parts. First, the notes will be described as they are published. Next, I will discuss how I used the notes as they were published in a two course upper level geometry sequence at the University of Northern Colorado. Following that, I will present how I adapted the two-course Modern Geometry I and II notes for use in a one semester course at Texas A&M University-Commerce. The review will conclude with a discussion of the strengths of the notes and a reflection on how my own teaching practice has become more student-centered through the use of these notes.

2 Summary of Published Notes

The Modern Geometry I and II notes are based on Henderson and Tamina's *Experiencing Geometry* [3]. The Modern Geometry I notes correspond with the material in chapters 1-4, 6, and 7 of that book. The notes may be taught with or without the companion textbook. The Modern Geometry I notes consist of thirteen assignments. Four of the assignments have been designated as formal assignments. These assignments are not more difficult than other assignments; rather, the formal assignments deal with fundamental geometric ideas: straightness, triangle congruence, symmetry proofs and a capstone project investigating the geometry of cones. The informal assignments deal with similar triangles, symmetries of quadrilaterals, Euclid's postulates on a sphere, neutral geometry—adapted from Coppin [7]—and proof, Euclid's postulates on a 450° cone, the area of a triangle on a sphere, angles of reflection found in a billiard table, an investigation into the NCTM standards, and the shape of space. This last activity asks students to conduct a thought experiment and use their knowledge of intrinsic geometry to determine what surface the universe might be. The Modern Geometry I notes also have four appendices. Appendix A contains feedback for common student misunderstandings, Appendix B contains an excerpt of *Flatland*, Appendix C is an excerpt of Euclid's *Elements* and Appendix D contains selected problems from Henderson & Tamina [3]. Appendix A in particular is a helpful resource for students as they write up solutions to the assignments and helps to expedite some of the paper commenting for me as an instructor. When I first started teaching from these notes, I found Appendix A helpful when I tried to plan which topics were most important to include in whole class discussions.

The Modern Geometry II notes correspond to chapters 5, 8, and 10 of

Table 1: Summary of Modern Geometry I and II notes

Modern Geometry I	Modern Geometry II
Viewing Tubes	Taxicab Geometry (Formal)
Straightness on the Sphere (Formal)	Create Hyperbolic Plane
Computer Lab I: Quadrilaterals	Euclid's Postulates on the Hyperbolic Plane
Euclid's Postulates on the Sphere	Non-Euclid
Neutral Geometry Worksheets	Folding a Parallel Line
Vertical Angle Theorem (Formal)	Regular Tessellations and Polyhedra
SAS on the Sphere (Formal)	Computer Lab I: Paper Folding
Euclid's Postulates on a 450° Cone	Computer Lab 2: The Product of Two Reflections
The Area of a Triangle on a Sphere	Parallel Postulates
Computer Lab II: Billiards Tables	Final Projects
NCTM Standards Assignment	
The Shape of Space	
Geometry on the Cone (Formal)	

Henderson and Tamina [3], and this semester of notes is intended to be taught alongside the text. There are only ten assignments in the second semester of notes, two of which have been designated as formal assignments. These assignments, which are the first and last assignments of the course, are similar to the cone project that ends the first semester of notes. In the first assignment, students are asked to prove as many things as they can about Taxicab geometry. The final formal assignment asks students to choose a chapter of Henderson and Tamina [3] not covered by any previous assignment and investigate the problems. I used presentations of the final formal assignment as a major component of the final exam when I taught this course using the notes as written. The informal assignments begin with three assignments introducing hyperbolic geometry (create a hyperbolic plane, Euclid's postulates on the hyperbolic plane, and Non-Euclid, which helps students see the different representations of the hyperbolic plane as equivalent). The other informal assignments include two in-depth parallelism assignments where students explore parallelism on the plane, sphere, and hyperbolic plane, two dynamic geometry computer labs where students are composing transformations, and one exploration of regular tessellations and polyhedra. As in the Modern Geometry I notes, Appendix A contains feedback for common student issues on the assignments, Appendix B contains excerpts of Euclid's Elements, Appendix C contains the relevant problems from Henderson and Tamina [3] for the informal assignments, and Appendix D is an excerpt from Douglas Hofstadter about the history of non-Euclidean geometry [4]. Table 1 provides a summary of the sequence of assignments for each set of notes.

3 Using the Notes in a Two Semester Geometry Course

At the first institution where I used the Modern Geometry course notes, the design of the program allowed the notes to be used as originally written. After describing the institution and course, I discuss student reactions to the IBL geometry courses and provide suggestions for implementing the notes as published.

3.1 Description of the Institution & Course

At the University of Northern Colorado, there were two semesters of college geometry, and I worked with the notes as originally published. In the fall semester I worked as a TA in a College Geometry I course and a College Geometry II course, while I was the main instructor in a spring College Geometry I course. In the College Geometry I course there were elementary education and secondary education majors with a few non-education majors taking the course as an elective. The second semester of college geometry was only required for secondary mathematics majors, so almost all students in the classes were secondary mathematics teachers.

The Geometry I class in the fall had 12 students; there were 26 students (an unusually large class) in Geometry II. The notes were taught as published with one exception. Neither of the Geometry I courses taught during the previous spring semester had completed the final informal assignment, so Geometry II began with The Shape of Space instead of the Geometry II notes. That semester, I was primarily a teaching assistant. My main role was to assist groups and grade papers. I also participated in a final cone project group in the Geometry I course. Each day during class, students were working in small groups of two to four students on one of the prompts in the notes. It normally took a week of class time before students had mastered enough about a prompt for the instructor to assign a write-up to the class. We prompted students to write a paper about everything they learned about the answer to the question posed in the notes, including pictures, data collection, examples, conjecture and proof like they were writing an explanation to a student. A paper was considered complete if students explained the process they took to arrive at a conjecture and the conjecture was correctly proven. For formal assignments, students were allowed to revise their paper once; informal assignments could be revised as many times as a student wanted. Most students took about three revisions to produce a complete solution.

While formal assignments were given letter grades, the informal assign-

ments were graded using a circle grading system based on a 12 point scale. Rather than a numerical grade, students received an arc of a circle that represented the completeness of their solution; for instance a paper with a half-complete solution would receive a half circle. To expedite grading, students turned in their original papers with their revisions. The informal assignments were 40% of the final grade and formal assignments were 60% of the grade.

3.2 Student Reactions

Although students were initially concerned by the non-standard grading system, they quickly accepted that a full circle was a 100% on the assignment; our expectations were that all students were eventually capable of earning full circles on all assignments. To help ease student nervousness during the first formal and informal assignments, we conducted extensive classroom discussions about expectations and had students formally peer review each other's papers before the first submission. Students mentioned this peer review in the student evaluations as one of things that helped them to adjust to the new course format and ease their anxiety.

3.3 Suggestions for Implementation of the Notes as Written

I have several suggestions for implementing the two semester sequence described in these geometry notes. First, depending on the size of the class, it may be helpful to allow groups to write collaborative papers on some informal assignments like the computer labs to ease grading workload. Second, the computer labs take a long time to implement, particularly the billiards table lab in Geometry I and the Paper Folding lab in the Geometry II notes. I never had enough time to implement the billiards table lab in Geometry I, and the paper folding activity can take up to two weeks of class time to implement. Since the final project needs 3-4 weeks of class time to implement, these assignments can be difficult to work into the end of the course. Third, it can be difficult to reach The Shape of Space without rushing the Geometry I notes, but this assignment makes a good review assignment at the start of Geometry II. At my current institution, I use an adaptation of The Shape of Space in math methods as a way to review geometric properties in the secondary mathematics methods course.

4 Adapting the Notes to a One Semester Geometry Course

At my current institution, there is only one upper level geometry course (non-Euclidean Geometry) for prospective secondary mathematics teachers. The other geometry course offered is primarily intended for prospective middle school majors and only covers basic Euclidean geometry. Although secondary mathematics education majors can take this course as an elective, the scope and sequence of the course is too divergent from the Modern Geometry notes for these notes to be an appropriate resource for that class. This necessitated adapting the two semester course to one that would fit into a 15 week semester. After describing the institution and changes I made to the course, student response to the course and suggestions for implementing the adapted model are discussed.

4.1 Description of the Institution & Course

Texas A&M University-Commerce is a rural research university located east of Dallas in the Texas A&M University system. There are two geometry courses offered, both of which are required for prospective secondary mathematics education majors. The first course (Introductory Geometry) is a course taken by pre-service middle school and occasionally high school teachers. The second course (Non-Euclidean Geometry) is required for secondary mathematics education majors and is a common elective that engineering and physics majors take to complete a second major in mathematics. Up to 33% of a given class is comprised of students taking the course as an elective. Class sizes range from 10-25 students, with most classes having about 20 students. Prospective high school teachers are encouraged to take Introductory Geometry first, but can take the two geometry classes in any order in their program.

The grading system and structure of the class at TAMUC is nearly identical to the one I used at UNCO, except groups are always randomly assigned at TAMUC. This is to ensure the education and non-education majors work with each other regularly, which they did not do the first semester I taught geometry at TAMUC and allowed students to choose their own groups. I randomly group students using sorting pencils (Figure 1).

Sorting pencils are a commercially available teaching aid that can be found at most teacher stores or through online sellers like Amazon. There are 36 pencils in the set. Each pencil has a color, shape, and number. You can form up to six groups of six using each of these characteristics as a grouping variable. To use these pencils in class, I begin by choosing a



Figure 1: Sorting pencils

grouping characteristic (such as shape or same parity + same color) and removing sufficient pencils so that I have the desired number of appropriately sized groups for the activity. Students are expected to draw a pencil from a jar at the front of the classroom whenever a new activity is started in class. After all students have drawn a pencil, I give the characteristic(s) to be used in forming groups for this activity.

During activities, students are in groups of 2-4, depending on the nature of the assignment and the size of the class. Since there is only one semester of Non-Euclidean geometry currently available to undergraduates, I selected a set of activities from both sets of notes to combine into a single semester Non-Euclidean Geometry course. Table 2 summarizes the activities used and their location in the original two semester sequence. For the one semester course, I chose to focus on students' understanding of geometric definitions and proving. The third formal assignment is typically given after Informal 6.

There are two major changes from the original published notes. Since Texas did not adopt the common core, students complete the 9th informal assignment with the Texas Essential Knowledge and Skills (TEKS) instead of the NCTM curriculum. Informal 2 in the original notes is a Geogebra lab where students use symmetries to classify quadrilaterals. Geogebra can have a steep learning curve for students, which can make this lab frustrating

Table 2: Condensed college geometry notes

Assignment	Original Assignment Number	Non-Euclidean Geometry
Informal 1	Geometry I Informal 1	Viewing Tubes
Formal 1	Geometry I Formal 1	Straightness on a Sphere
Informal 2	New Assignment	Shape Sorting
Informal 3	Geometry I Informal 3	Euclid's Postulates on a Sphere
Informal 4	Geometry II Informal 2	Make a Hyperbolic Plane (Take Home)
Formal 2	Geometry I Formal 3	SAS on a Sphere
Formal 3	Geometry I Formal 4	Midterm (Take Home)
Informal 5	Geometry I Informal 4	NG worksheets
Informal 6	Geometry II Informal 4	Euclid's Postulates on a hyperbolic plane
Informal 7	Geometry I Informal 6	Area of a triangle on the sphere
Informal 8	Geometry II Informal 5	Parallel Activity I
Informal 9	Geometry I Informal 8	Geometry TEKs (Take Home)
Informal 10	Geometry II Informal 9	Parallel Activity II
Informal 11	Geometry I Informal 9	The Shape of Space (optional)
Formal 4	Geometry I Formal 5	Cone Final Project

for my students. After two iterations of the course, I replaced Informal 2 with an activity based on Vendeira & Coutat [1]. Students have to create a taxonomy for a set of 36 mostly concave shapes (Figure 2). Students must create their own definitions for all categories in their taxonomy, and the polygons are on circular flashcards so that orientation cannot be a sorting criterion. The activity is considered completed for full credit once a group that has not seen the taxonomy before can successfully sort the shapes in the way the authors intend. The full activity appears at the end of this review as an appendix.

4.2 Student Response to the Course

The reaction to the IBL geometry notes has been positive from my pre-service teachers (PST); students are typically enrolled in an IBL in the secondary schools course the same semester they take college geometry. Our PST also take a traditionally taught Euclidean geometry course for middle school teachers before college geometry, and have mentioned that they like seeing the same material presented in two very different ways. Students taking college geometry as an elective have been more reluctant to embrace the inquiry-based nature of the course, but a discussion at the beginning of the semester about how inquiry is a part of all majors staves off most students' objections. I also allow non-education majors to complete an alternate in-

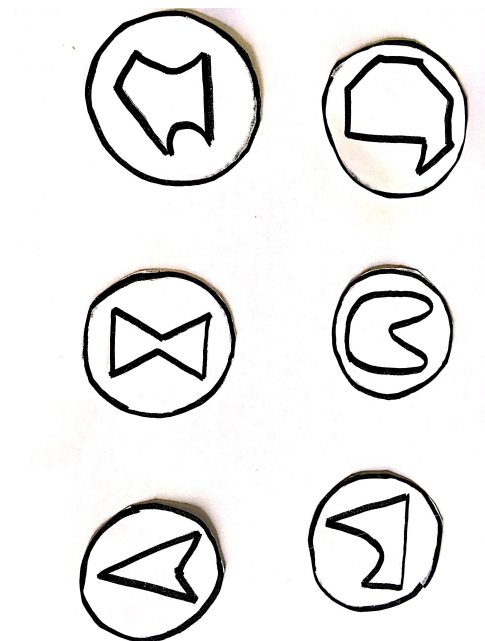


Figure 2: Sample shape cards

formal 9 assignment based upon modernizing Flatland instead of evaluating the TEKs. Although students are often concerned about the amount of writing involved in the class, this concern tends to go away if they are allowed to have a peer review on the first assignment.

4.3 Suggestions for Teaching Adapted Notes

When teaching the condensed version of the notes, I suggest to plan on roughly one week of class time per assignment (about 150 minutes) with three exceptions. The Non-Euclidean Geometry (NG) worksheets take 2.5–3 weeks and Informal 9 generally does not require class time. The final formal assignment should take 3–4 weeks of class time. If the NG worksheets take 3 weeks, Informal 10 may need to be shortened or eliminated in order to have sufficient time for the final project. If a class is particularly quick throughout the semester, I suggest adding the Shape of Space (Geometry I, Informal 9) or Taxicab Geometry (Semester 2, Formal 1) to the course as an additional informal assignment. I have 11 assignments listed under the informal assignments in case I need it on the course schedule in the syllabus, but only plan on 10 informal assignments during a typical semester.

5 Strengths of the Modern Geometry Notes

This freedom to explore conical geometry at the end of the semester, which models actual mathematical research, is the greatest strength of the Modern Geometry Notes. There are tradeoffs with the time commitment needed for such explorations though. In the condensed version of the notes, students do not spend any significant time working on Geogebra or other dynamic geometry software except for the cone projects. Instead, students use patty paper (planes), transparencies (spheres), or Non-Euclid (hyperbolic planes) to physically perform transformations. This is somewhat more expensive in terms of material cost, but avoids most of the learning curve involved with Geogebra. The second potential criticism of the notes is more one of philosophy. The activities in these college geometry notes are not as formal an axiomatic geometry proof class as a more traditional college geometry course, and Euclidean geometry receives relatively little emphasis. I do not find this particularly problematic since students may take a more traditionally-taught semester of Euclidean geometry as one of their electives. This ensures that our pre-service teachers have a strong grounding in the geometry content they could teach, and aligns well to the current AMTE teacher preparation standards for two semesters of geometry, one of which is non-Euclidean [2]. If I was using these notes for the only course offering, I would suggest de-emphasizing the hyperbolic geometry component in favor of the Geogebra labs to keep a more planar geometry feel to the course.

6 How the Modern Geometry Notes Influenced my Teaching

Like many instructors new to IBL, I found helping and grading when I first starting teaching from these materials as a TA more challenging than expected. It was difficult to ask questions that led students without pushing them to the correct solution, and grading took much more time than expected. Despite the steep learning curve, I could see that students were more engaged and thinking more deeply than they were in the traditionally taught proof-based classes I had taught in the past. By the end of the semester I had improved on grading enough to be confident in the new system. During the spring semester, I taught a ten student Geometry I course. By the end of the year I was convinced that students learned more deeply in an IBL environment. Overall, these geometry notes were a good place in the curriculum to experiment with IBL learning. Unlike calculus, geometry is rarely part of a sequence, and since the class is often populated by education majors, the

student population may be more open to non-traditional teaching methods.

These geometry notes were my first experience with teaching an IBL college course. Before these notes, I taught very traditionally. After observing IBL geometry then teaching it myself three times, I have shifted my teaching to a much more inquiry-oriented philosophy. Even though college geometry is the only course that I teach in a completely inquiry-based mode, I have incorporated the idea of productive struggle into all of my other classes in at least one project. The biggest shift in my teaching has occurred in calculus. Although I cannot teach a wholly IBL course due to curriculum constraints, I use a weekly IBL lab to supplement calculus instruction and help students develop an understanding that calculus concepts are based upon the idea of a limit. I am a more effective instructor in college geometry as well as my other courses for using these notes. I highly recommend these notes as a first experience a first experience in teaching an IBL or college geometry course.

References

- [1] Vendeira, C., & Coutat, S. (2017, February). Shape recognition in early school: how to develop the dimensional deconstruction? Paper presented at the 10th Congress of European Mathematics Education, Dublin, Ireland.
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- [4] Hofstadter, D. (1999). *Godel, Escher, Bach: an Eternal Golden Braid, 20th Anniversary Edition*. New York: Random House.
- [5] Miller, N. (2010a). Modern Geometry I course notes. *Journal of Inquiry-Based Learning in Mathematics*, 17, 1–62.
- [6] Miller, N. (2010b). Modern Geometry II. *Journal of Inquiry-Based Learning in Mathematics*, 19, 1–46.
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Appendix: Taxonomy

Informal Assignment 2: Shape Sorting

(Suggested group size: 2 students/group)

Student prompt in coursepack:

Taxonomies are classification schemes that identify objects or concepts by properties. These schemes are generally partitions, which mean that each object belongs to one and only one category. For this assignment, your group will create a taxonomy of the 36 shapes on the cards passed out to you. You must create your own definitions for all categories and may not use any standard geometry terms to define your categories. Your taxonomy is successful if it is sufficiently descriptive if someone who has only read your description of the categories can correctly sort all 36 shapes.

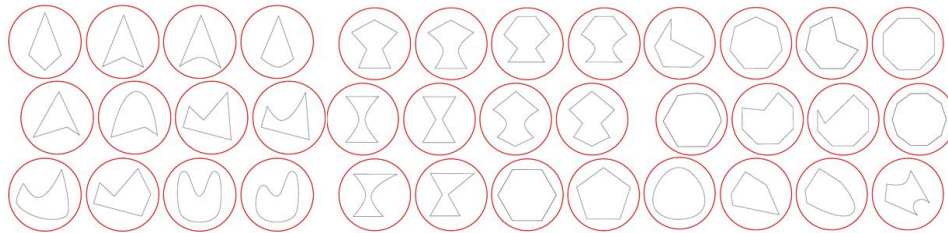


Figure 3: The shape cards [1]

Instructor notes:

If students seem really stuck, I will prompt them to try and come up with 5-8 categories. Star Trek Communicator (e.g. top row, second from the left) is a common category, as is Fish (e. g. bottom right) or Pacman (top row, 2nd from right). There is generally also considerable debate where or not shapes with all straight sides can be included with shapes with sides that curve in the same category. The shapes have deliberately been placed on round cards so that students may not rely on the orientation of the card as a way to sort the shapes. Students typically have the most trouble including the 4th shape from the right on the bottom row into a category, and may occasionally have it in its own category. I strongly recommend that groups take a picture of their shapes sorted into groups before allowing another group to attempt to sort the shapes using their taxonomy. Although students

are initially extremely confident that their taxonomy is unambiguous and easy to sort, groups typically need to have 3-4 cycles where other pairs of students attempt to sort according to their taxonomy and then revising and clarifying their definitions before a group can successfully sort the shapes as intended. For the written assignment, I ask each pair to turn in their written taxonomy and a picture of the shapes sorted as intended. Full credit is awarded if the instructor can successfully sort the shapes.